6.1 Purpose of Testing

Testing checks whether the electricity billing program produces correct results for different inputs. It helps identify errors in meter readings, rollover calculations, tariff calculations, and total bill amounts.

6.2 Test Cases

1. Zero units: Verify that the fixed charge is applied when consumption is zero.

2. Exactly 100 units: Verify that the first 100 units are free of energy charges.

3. 150 units: Verify that 50 units are charged at the second tariff rate.

4. 200 units: Verify that the correct tariff is applied up to 200 units.

5. 350 units: Verify that the correct rates are applied to both tariff slabs.

6. Meter rollover: Verify that a previous reading of 65400 and a current reading of 120 produce 256 units.

7. Invalid readings: Verify that invalid meter readings are rejected.

6.3 Expected Results

The expected bill amounts depend on the defined tariff rates and fixed charge. For the example rates used in this assignment, the fixed charge is ₹50, the second tariff is ₹2.25 per unit, and the third tariff is ₹4.50 per unit.

6.4 Alternative Concept and Complexity Analysis

Manual testing checks selected examples, while automated testing runs multiple test cases consistently. Testing (n) cases takes (O(n)) time. Each test requires constant additional memory, so the extra space per test is (O(1)).

Conclusion

Testing helps ensure that the billing program handles normal consumption, tariff boundaries, meter rollover, and invalid inputs correctly.

```
#include <stdio.h>

long compute_bill(unsigned int units)
{
    long paise = 5000;

    if (units > 100 && units <= 200)
        paise += (long)(units - 100) * 225;
    else if (units > 200) {
        paise += 100L * 225;
        paise += (long)(units - 200) * 450;
    }

    return paise;
}

int main(void)
{
    unsigned int tests[] = {0, 100, 150, 200, 350};
    int n = sizeof(tests) / sizeof(tests[0]);

    for (int i = 0; i < n; i++) {
        long bill = compute_bill(tests[i]);

        printf("%u units: Rs %ld.%02ld\n",
```

```
            tests[i], bill / 100, bill % 100);
    }

    printf("Rollover consumption: %u units\n",
           (unsigned short)(120 - 65400));

    return 0;
}
```

